

 Marwadi University Marwadi Chandarana Group		Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulation of Cross-Correlation and Auto-Correlation on Discrete-Time Signals	
Experiment No: 3	Date: 02/07/2026	Enrollment No: 92400133147

Aim

To simulate the auto-correlation and cross-correlation of two discrete-time signals using Python, analyze their similarity, and compare the correlation characteristics through graphical representation.

Objectives

1. To understand the concept of correlation in Digital Signal Processing (DSP).
2. To generate two discrete-time signals.
3. To compute the auto-correlation of each signal.
4. To compute the cross-correlation between the two signals.
5. To compare the auto-correlation characteristics of both signals.
6. To compare the cross-correlation characteristics of both signals.
7. To visualize the original signals and their correlation plots.
8. To understand the applications of correlation in signal processing.

Theory

Correlation is a mathematical operation used to measure the similarity between signals. It is widely used in Digital Signal Processing for pattern recognition, signal detection, synchronization, and feature extraction.

Auto-Correlation

Auto-correlation measures the similarity of a signal with a delayed version of itself. It helps identify periodicity and repeating patterns in a signal.

The auto-correlation of a discrete-time signal is given by:

$$R_{xx}(k) = \sum_{n=-\infty}^{\infty} x(n)x(n-k)$$

where:

- $x(n)$ = Original signal
- k = Time lag

The maximum value of the auto-correlation occurs at zero lag.

Cross-Correlation

Cross-correlation measures the similarity between two different signals as one signal is shifted with respect to the other.

The cross-correlation is expressed as:

$$R_{xy}(k) = \sum_{n=-\infty}^{\infty} x(n)y(n-k)$$

where:

- $x(n)$ = First signal
- $y(n)$ = Second signal
- k = Time lag

A larger correlation value indicates greater similarity between the two signals.

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Software Requirements

- NumPy
- Matplotlib
- VS Code

Input

Two discrete-time signals:

Signal 1

$$x(n) = \{1, 3, 2, 5, 4, 2, 1\}$$

Signal 2

$$y(n) = \{2, 1, 4, 3, 5, 2, 1\}$$

Code and Output:

```
import numpy as np
import matplotlib.pyplot as plt
import os
```

[1] ✓ 0.8s

```
output_folder = "Correlation_Images"

os.makedirs(output_folder, exist_ok=True)

print("Folder Created Successfully")
```

[2] ✓ 0.0s

... Folder Created Successfully

```
signal1 = np.array([1, 3, 2, 5, 4, 2, 1])

signal2 = np.array([2, 1, 4, 3, 5, 2, 1])

print("Signal 1 =", signal1)
print("Signal 2 =", signal2)
```

[3] ✓ 0.0s

... Signal 1 = [1 3 2 5 4 2 1]
Signal 2 = [2 1 4 3 5 2 1]

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```
plt.figure(figsize=(8,4))

plt.stem(signal1)

plt.title("Original Signal 1")

plt.xlabel("Sample")

plt.ylabel("Amplitude")

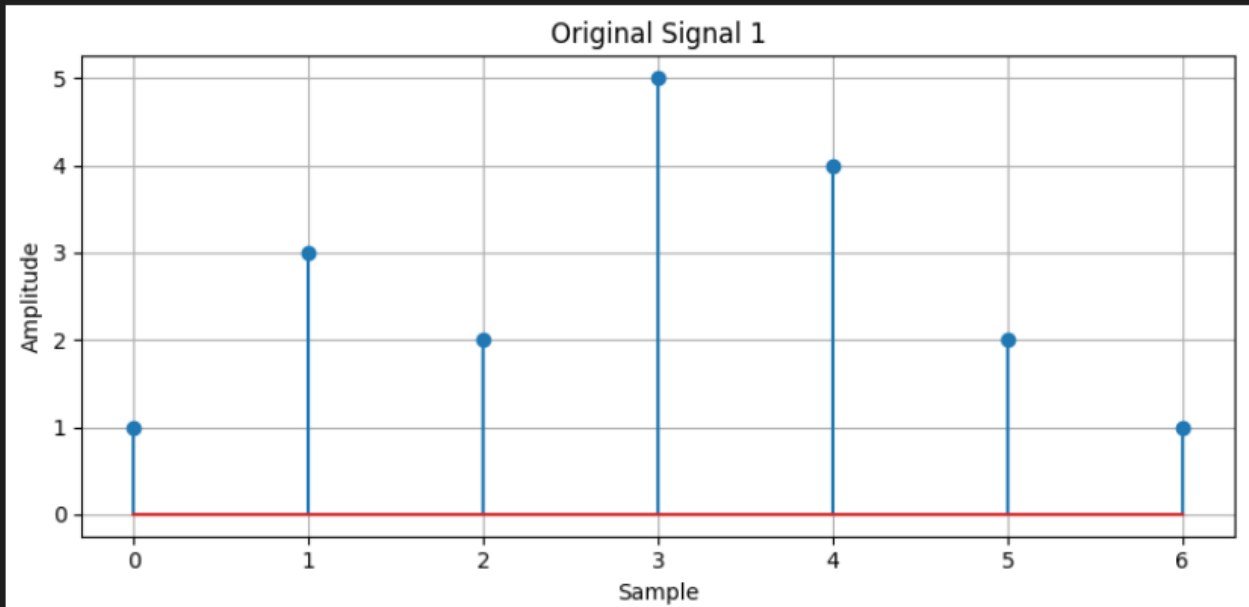
plt.grid(True)

plt.tight_layout()

plt.savefig(os.path.join(output_folder,"Original_Signal1.png"))

plt.show()
```

[4] ✓ 0.1s



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```

plt.figure(figsize=(8,4))

plt.stem(signal2)

plt.title("Original Signal 2")

plt.xlabel("Sample")

plt.ylabel("Amplitude")

plt.grid(True)

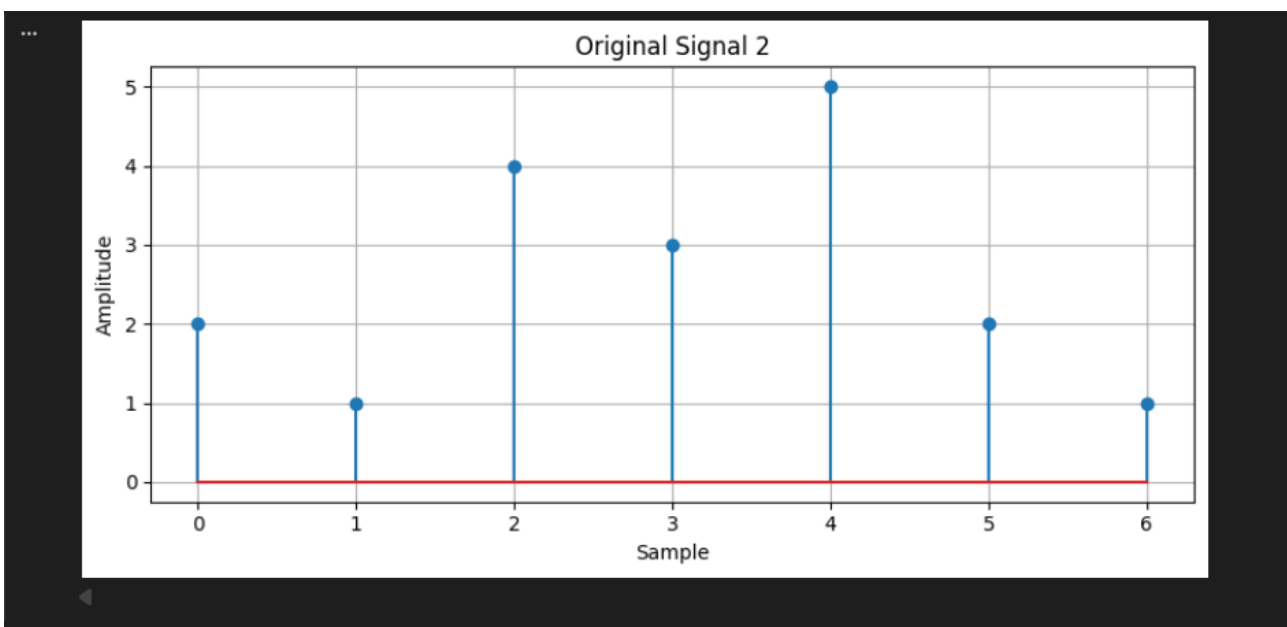
plt.tight_layout()

plt.savefig(os.path.join(output_folder,"Original_Signal2.png"))

plt.show()

```

[5] ✓ 0.1s



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```

auto1 = np.correlate(signal1, signal1, mode="full")

lags1 = np.arange(-(len(signal1)-1), len(signal1))

plt.figure(figsize=(8,4))

plt.stem(lags1, auto1)

plt.title("Autocorrelation of Signal 1")

plt.xlabel("Lag")

plt.ylabel("Correlation")

plt.grid(True)

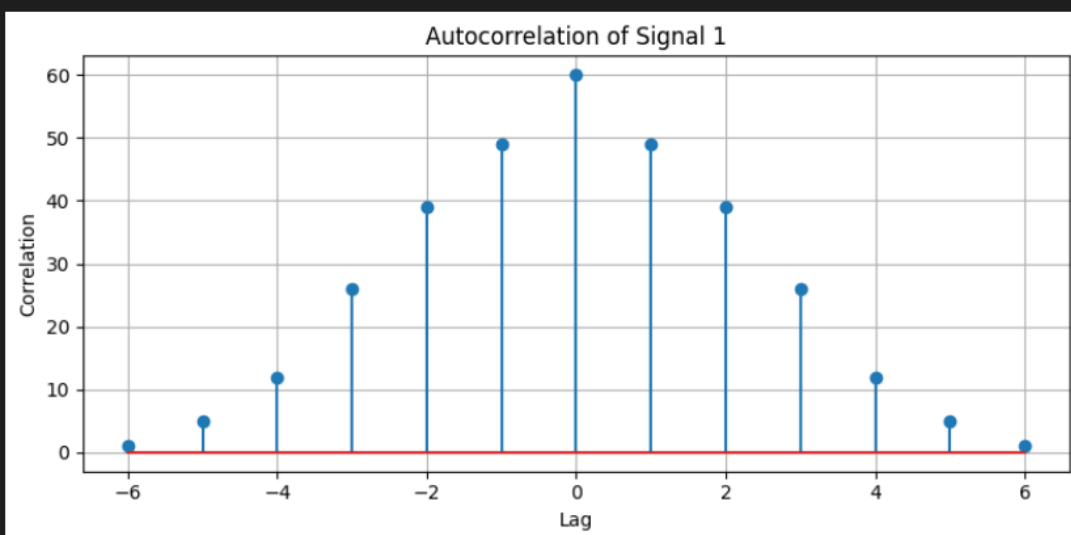
plt.tight_layout()

plt.savefig(os.path.join(output_folder,"Autocorrelation_Signal1.png"))

plt.show()

```

✓ 0.1s



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```

auto2 = np.correlate(signal2, signal2, mode="full")

lags2 = np.arange(-(len(signal2)-1), len(signal2))

plt.figure(figsize=(8,4))

plt.stem(lags2, auto2)

plt.title("Autocorrelation of Signal 2")

plt.xlabel("Lag")

plt.ylabel("Correlation")

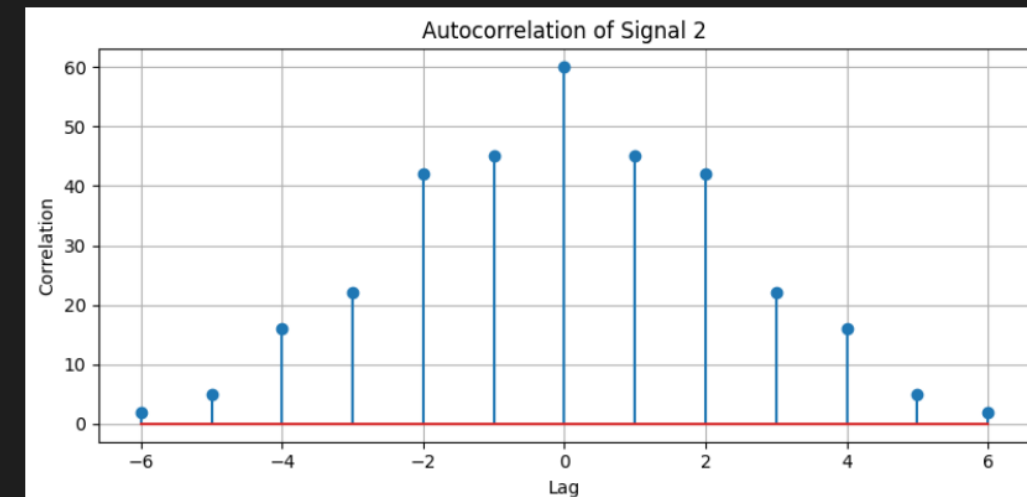
plt.grid(True)

plt.tight_layout()

plt.savefig(os.path.join(output_folder,"Autocorrelation_Signal2.png"))

plt.show()

```



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```

cross12 = np.correlate(signal1, signal2, mode="full")

lags12 = np.arange(-(len(signal2)-1), len(signal1))

plt.figure(figsize=(8,4))

plt.stem(lags12, cross12)

plt.title("Cross Correlation of Signal 1 with Signal 2")

plt.xlabel("Lag")

plt.ylabel("Correlation")

plt.grid(True)

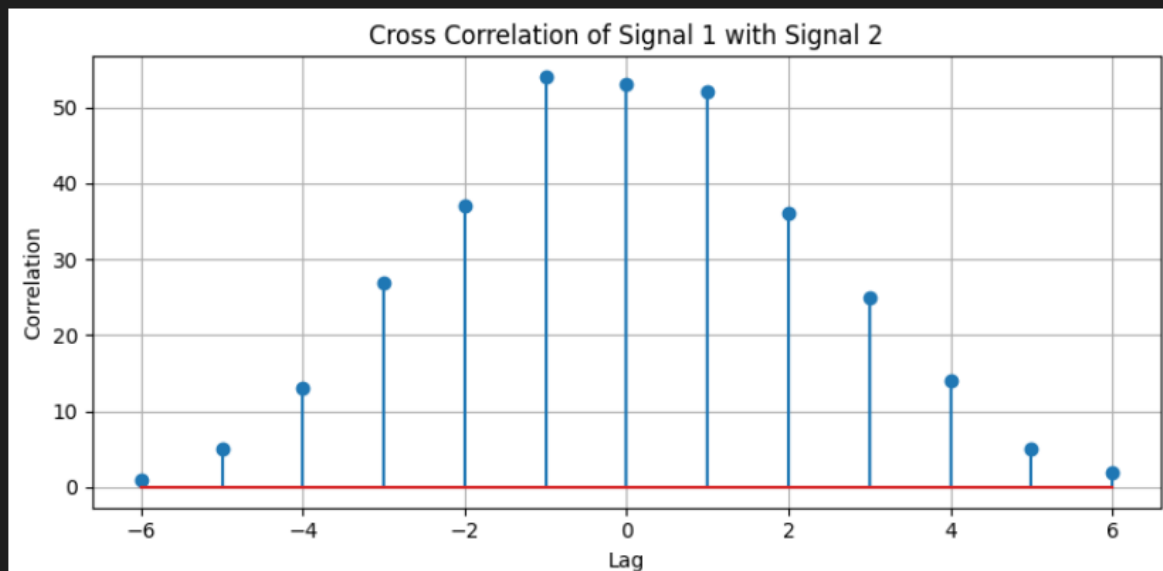
plt.tight_layout()

plt.savefig(
    os.path.join(
        output_folder,
        "CrossCorrelation_Signal1.png"
    )
)

plt.show()

```

✓ 0.2s



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```

cross21 = np.correlate(signal2, signal1, mode="full")

lags21 = np.arange(-(len(signal1)-1), len(signal2))

plt.figure(figsize=(8,4))

plt.stem(lags21, cross21)

plt.title("Cross Correlation of Signal 2 with Signal 1")

plt.xlabel("Lag")

plt.ylabel("Correlation")

plt.grid(True)

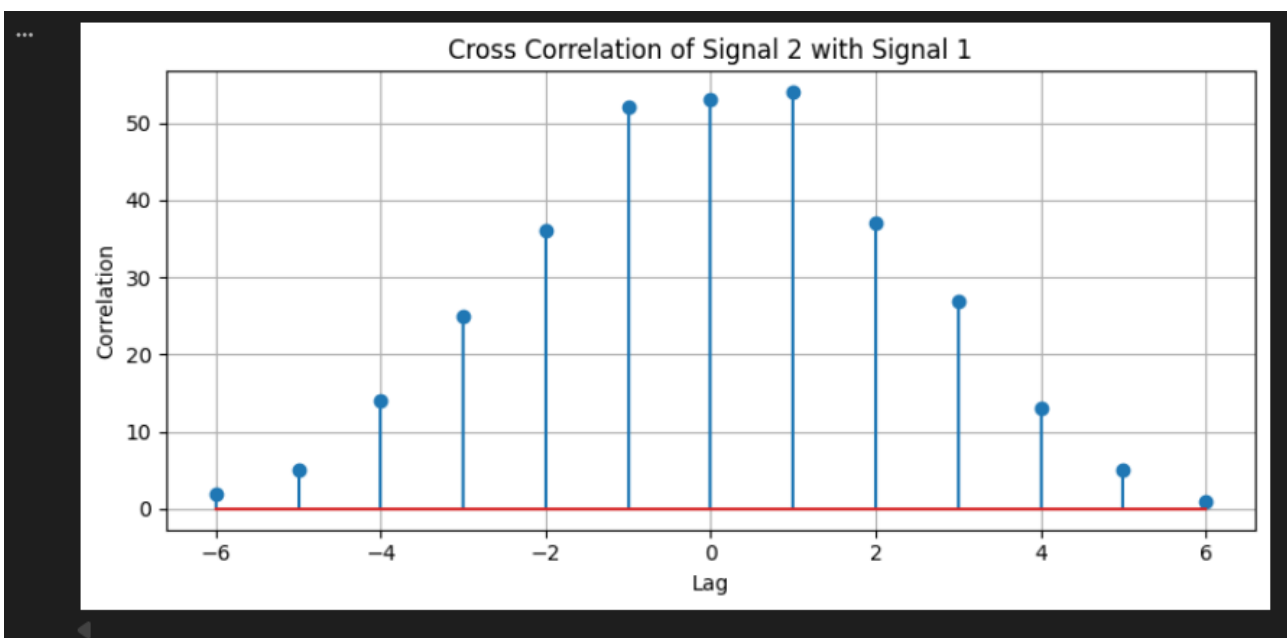
plt.tight_layout()

plt.savefig(
    os.path.join(
        output_folder,
        "CrossCorrelation_Signal2.png"
    )
)

plt.show()

```

[9] ✓ 0.2s



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```

plt.figure(figsize=(10,5))

plt.plot(lags1, auto1,
         marker='o',
         linewidth=2,
         label="Signal 1")

plt.plot(lags2, auto2,
         marker='s',
         linewidth=2,
         label="Signal 2")

plt.title("Comparison of Auto Correlation")

plt.xlabel("Lag")

plt.ylabel("Correlation")

plt.legend()

plt.grid(True)

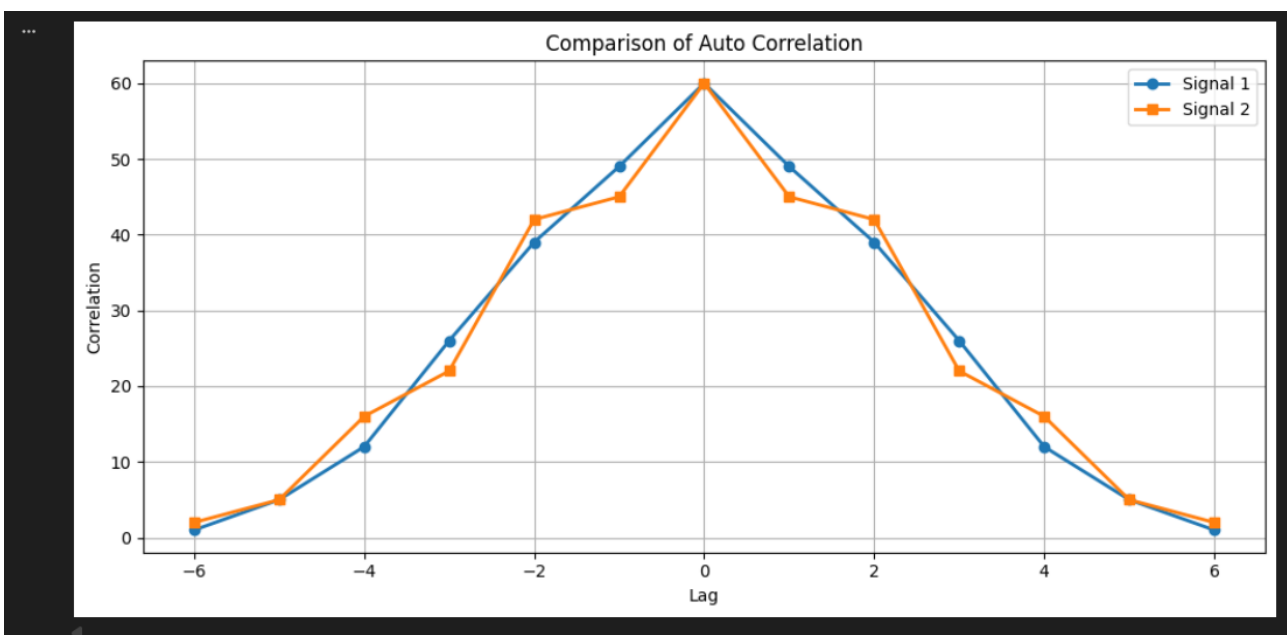
plt.tight_layout()

plt.savefig(
    os.path.join(
        output_folder,
        "AutoCorrelation_Comparison.png"
    )
)

plt.show()

```

[10] ✓ 0.2s



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```

plt.figure(figsize=(10,5))

plt.plot(lags12, cross12,
         marker='o',
         linewidth=2,
         label="Signal1 vs Signal2")

plt.plot(lags21, cross21,
         marker='s',
         linewidth=2,
         label="Signal2 vs Signal1")

plt.title("Comparison of Cross Correlation")

plt.xlabel("Lag")

plt.ylabel("Correlation")

plt.legend()

plt.grid(True)

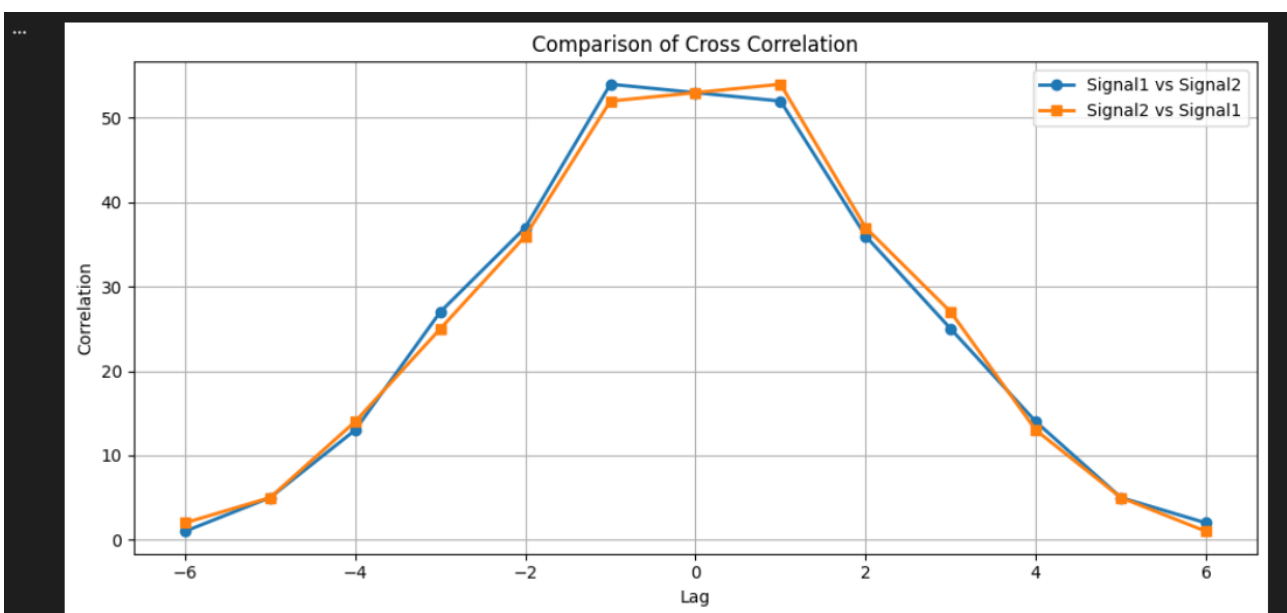
plt.tight_layout()

plt.savefig(
    os.path.join(
        output_folder,
        "CrossCorrelation_Comparison.png"
    )
)

plt.show()

```

[12] ✓ 0.2s



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Output

The program generates eight images:

1. Original Signal 1
2. Original Signal 2
3. Auto-Correlation of Signal 1
4. Auto-Correlation of Signal 2
5. Cross-Correlation of Signal 1
6. Cross-Correlation of Signal 2
7. Comparison of Auto-Correlation
8. Comparison of Cross-Correlation

Observations

Original Signal 1

The first discrete-time signal consists of varying amplitudes and is used as the reference signal.

Original Signal 2

The second signal has a different amplitude pattern, allowing comparison through cross-correlation.

Auto-Correlation of Signal 1

The highest peak occurs at zero lag, indicating perfect similarity with itself.

Auto-Correlation of Signal 2

The maximum correlation also occurs at zero lag, confirming self-similarity.

Cross-Correlation of Signal 1

The graph shows the similarity of Signal 1 with shifted versions of Signal 2. The maximum peak represents the highest similarity.

Cross-Correlation of Signal 2

The graph represents the similarity of Signal 2 with shifted versions of Signal 1. The peak occurs at a different lag due to the reversed order of comparison.

Comparison of Auto-Correlation

Both signals exhibit symmetric auto-correlation graphs with maximum values at zero lag. Differences in amplitude arise from the different signal values.

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Comparison of Cross-Correlation

The two cross-correlation graphs contain similar correlation values but appear as reversed versions of each other because the order of the signals is interchanged.

Advantages

- Simple to implement.
- Measures signal similarity accurately.
- Useful for periodic signal detection.
- Helpful in synchronization and delay estimation.
- Widely used in communication and radar systems.

Limitations

- Computational complexity increases for long signals.
- Correlation values may be affected by noise.
- Cross-correlation depends on proper signal alignment.
- Large datasets require greater computation time.

Result

The auto-correlation and cross-correlation of two discrete-time signals were successfully simulated using Python. Eight graphical outputs were generated, including the original signals, individual auto-correlation plots, individual cross-correlation plots, and comparison graphs. The results clearly demonstrate the similarity characteristics of the signals at different time lags.

Conclusion

The experiment successfully demonstrated the implementation of auto-correlation and cross-correlation for two discrete-time signals. Auto-correlation showed that each signal is perfectly correlated with itself at zero lag, while cross-correlation measured the similarity between the two different signals at various time shifts. The comparison graphs highlighted the symmetric nature of auto-correlation and the reversed relationship between the two cross-correlation results. This experiment illustrates the importance of correlation techniques in digital signal processing applications such as synchronization, pattern recognition, signal detection, and communication systems.

GitHub:

<https://github.com/Devaharshagoud/DSIP-Experiments.git>